



Improving demand responsive transit services: Insights from the London field test

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ABSTRACT

The paper discusses the key insights gained from “GoSutton”, a Demand Responsive Transit (DRT) trial in the borough of Sutton in the suburbs of London during 2019–2020. The DRT service expands mobility options for the residents of the service area through the provision of on-demand, curb-to-curb rides. The main goals of the paper are to: 1) evaluate the performance of the DRT service and provide insight into its operation, 2) analyze the behavior of customers to better understand their purchase decisions and recommend strategies to increase ridership and usage, 3) develop a Level of Service (LOS) model to inform future trials on the interactions between supply, demand, and LOS, as well as identify key factors that impact performance, and 4) evaluate the opportunities to benefit from economies of scale through integrating appropriate paratransit requests and DRT services. The results show potential directions for improvement that can inform future deployments (for example, providing discounts for referrals, new users, providing bundles, and passes). Consolidation of select paratransit and DRT trips can save an average of 8 % on fleet size and 13 % on vehicle-kilometers traveled.

Introduction

In rural and suburban areas users of conventional public transport services are confronted with low accessibility and a lack of transit choices (Schasché and Sposato, 2021). Spatial and temporal sparsity of travel demand in rural and suburban areas affects the efficacy and cost-effectiveness of conventional transit operations (Tilahun et al., 2016). Demand-Responsive Transit (DRT) is considered as a potential solution to these issues. DRTs are flexible in their schedules and routes and can provide demand-oriented services that focus on passenger's needs. DRT vehicles are usually smaller in size than traditional buses and are not constrained by a timetable which enables them to react to users' travel patterns. A formal definition of DRT services can be found in (Ambrosino et al., 2004) which places DRTs somewhere between buses and taxis as an intermediate form of transport. DRT is a shared transit service where vehicles can modify their routes and schedules at each run based on a particular realization of transit demand.

Although DRT was first introduced in the 1970s, there is currently a renewed interest in DRT services among cities and transit authorities (Liyanage et al., 2019). Various reasons contribute to the rising popularity of DRT (Laws et al., 2009). Recent advances in communication

technology and paperless payments have resolved many obstacles that hindered the use of DRTs in the past. Smartphone apps have made it easier and safer for riders to request a ride. Recognition of the fact that if conventional transit does not adapt to changing conditions (both in terms of customer expectations, and opportunities provided by technological advances) will negatively affect its competitiveness, is increasingly convincing transit authorities to pursue deployments of DRT services.

DRT services can differ significantly in their operation and design. Koffman categorized DRT services into six main types, among which, route-deviation services, flexible-route segments, and Demand-responsive connectors (DRCs) are the most prevalent types (Koffman, 2004). Route-deviation services allow transit vehicles to deviate from their pre-defined path to serve customers in an on-demand fashion. Flexible route segments resemble route-deviation services, but the deviation only occurs on specific segments along their routes. DRCs are many-to-one and one-to-many services that are usually used as feeders to major transit stations. Hybrid designs that combine different types of DRT with each other or with conventional transit have also been proposed in the literature (Aldaihani et al., 2004).

In addition to the differences in design and operation, there are also

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different use cases and applications of DRTs. Most DRT operations pertain to paratransit services that offer door-to-door rides to residents with disabilities. Although Paratransit operations address an important need, they are traditionally cost-inefficient, currently making up more than 12.2 % of transit costs in the United States (Puentes et al., 2016). Other than paratransit, DRT services have three main applications: First/Last Mile services, bus route replacements, and expansion of mobility options (Koutsopoulos et al., 2023).

First/Last Mile services. DRT services are used to provide first-mile or last-mile access connecting riders to public transport options. The aim is to increase the catchment area of transit lines, increase ridership, and shift travel mode choices away from cars to public transportation through multi-modal trips. Go Centennial in Colorado, BerlKönig in Berlin, and BRIDJE in Sydney are examples of such services (Perera et al., 2020).

Bus route replacement. When travel demand density is low, DRT can potentially provide a better Level of Service (LOS) with lower costs (Quadrifoglio and Li, 2009). DRT services can improve accessibility by providing curb-to-curb service and increase efficiency through optimal matching of riders and vehicles. Curb-to-curb service contrasts door-to-door service. Trips do not start and end at the exact location of the users but at designated locations that are close and within walking distance (1–2 minutes) from the user's desired locations. The cities of Innisfil, Ontario, and Arlington, Texas partnered with Uber and Via respectively to replace some of their underperforming bus services with demand responsive services which were very popular with the residents of those cities (Schwieterman et al., 2018).

Expanding mobility options. Recently DRT services are also viewed as a complementary mode of travel providing additional mobility options to riders. These DRT services operate in addition to and complementary to other modes of public transportation to provide riders with a variety of options to choose from as they see fit. Kutsuplus in Helsinki, GoSutton, and Slide Ealing in London are examples of such applications of DRT (Haglund et al., 2019).

From a financial point of view, DRT services are prone to failure. Currie and Fournier report that 25 % of DRT services fail within two years and 40 % last less than 3 years (Currie and Fournier, 2020). 67 % of DRTs in the UK and 54 % in Australia have failed in the last 40 years. Various reasons are identified for the failures of DRT services in the literature. The most prominent are high costs in relation to their patronage and lack of understanding of the DRT market (Enoch et al., 2006). DRT trials offer a good opportunity to better understand operations and customer demand characteristics and to identify ways to increase ridership and reduce costs.

The objective of this paper is to outline the key insights gained from the "GoSutton" DRT field test in London, UK, and use the findings to inform future deployments of such services. The insights revolve around the factors that impact the performance of the service, the behavior of the customers, and opportunities for taking advantage of economies of scale. More specifically, the paper addresses the following questions:

- What are the major factors that impact the performance of the service?
- How can agencies promote service and incentivize users, especially in light of opportunities provided by the mobile internet and social media?
- How to take advantage of economies of scale to increase the efficiency of the service?

With the aforementioned objectives, the contributions of the paper are as follows:

- An in-depth discussion of the performance of the system and its operating characteristics.
- A model that predicts LOS as a function of demand, supply, and other explanatory variables.
- Better understanding of customer behavior, analyzed over time, through novel metrics adopted from the marketing literature and

used by companies to increase customer loyalty and product demand. This understanding can be used to inform strategies that may increase DRT patronage.

- Identification and exploration of opportunities to improve operating efficiency taking advantage of economies of scale through consolidation of (select) paratransit trips and DRT trips.

The rest of the paper is organized as follows: Section 2 explores various recent global implementations of DRT services, their characteristics, and key takeaways, drawing insights from a wide range of studies and examples. Section 3 reports on the background of the trial and the operation. Section 4 discusses the general characteristics of the service. Section 5 analyzes the LOS of the service. Section 6 explores the behavior of the customers, examines useful indicators, and provides recommendations to increase usage and promote loyalty. Section 7 discusses consolidation strategies to take advantage of economies of scale. Section 8 concludes the paper.

Literature review

In Europe, one recent example of DRT services is a DRT pilot in Helsinki, Finland operated by Viavan. The pilot demonstrated the complementary role of the urban DRT to the network of fixed-route public transit. Further research on this pilot developed the concept of phase-based time-scale to help understand the service trajectory of DRT service over time. In this framework, the phases are based on key changes in service design (Saif et al., 2024). Another example in Bern, Switzerland, a DRT scheme (EBuxi) was developed as a first-/last-mile solution providing a door-to-door, shared transportation on demand with flexible routes and flexible schedules. The service was initiated and operated by a private actor and a voluntary organization. Studies on Ebuxi in the peri-urban area reveal that DRT services in such niche markets can reduce operating costs. They also found that users are more tolerant of wait times than detour times (Thao et al., 2023). In Vulkaneifel, Germany, a study focused on the feasibility of serving students with DRT and showed that indeed DRT can provide an efficient school transport service with reasonable cost. They also showed that costs can be reduced by 35 % if bookings are only accepted in advance. The cost can be further reduced if the autonomous fleet is used for the service (Lu et al., 2023).

In the United States, the Pickup service in Austin, Texas, serves as an insightful case study for assessing the impact of DRT on first-mile and last-mile connectivity. Managed in collaboration with the local transit authority, Pickup has proven effective in boosting ridership and enhancing connectivity, though it encounters difficulties with demand prediction and scheduling. The "Via to Transit" service in Seattle is an example of DRT deployment, offering flexible links to existing transit routes. The results also demonstrated the cost advantages of incorporating on-demand shared modes in areas where running fixed-route transit is costly (Wang and Shen, 2023).

In Asia, a study focusing on successful examples of DRT in Dalian, China, found the significant impact of the built environment on DRT usage. The study suggests that the built environment conducive to DRT success includes a larger residential population and more residential land uses, less administrative land uses, lower employment density, poor connectivity to road networks and parking supply, and long distances from urban business centers. These findings suggest a different built environment than the ones commonly known as appropriate for regular transit and can inform future DRT operations (J. Wang et al., 2023a, 2023b). In Dubai, UAE, the Dubai Bus on Demand (DBOD) has been operated by a third-party service provider that charges a distance-based fee. The study suggests that although having a user-friendly service with flexible timing and extra capacity for peak times is an important factor for DRT success, the cost recovery ratio of the service operation is the single most significant indicator for the commercial sustainability of a DRT system. Other factors such as marketing, technology, and driver

behavior also play a role in the success of the system (Deka et al., 2023).

Understanding customer behavior is crucial for optimizing DRT services. Recent research has highlighted that convenience, affordability, and reliability are key factors influencing user satisfaction (Jaber et al., 2023). A study on two DRT systems in Austria evaluated the relationship between customer satisfaction for different population groups and trip-level and system level attributes and confirmed that commonly known factors such as wait time impact customer satisfaction however, they found no negative relationship between satisfaction and the presence of other riders in a ride (Gkavra et al., 2024). Dynamic pricing strategies and compensation mechanisms are developed as a means to attract and keep users in DRT services (H. Wang et al., 2023a, 2023b). By adjusting prices according to demand, dynamic pricing can enhance service usage and user satisfaction, but it requires careful application to prevent adverse impacts on accessibility (Chen et al., 2023).

Some recent studies evaluated the energy impacts of DRT and the uses of new technologies such as autonomous vehicles in DRT operations. A case study in Arlington, TX, utilized mobility energy productivity (MEP) metric, to analyze and compare various transportation modes. Analyzing six scenarios, the results revealed that DRT improved the average MEP score across the service area by 83 % when considering only accessible, nonprivate vehicle modes (Powell et al., 2024). The National Renewable Energy Laboratory used the same metric across Dalas, Tx, Innisfil and Fort Erie, ON showed that DRT services are more time-efficient for users and more energy efficient in operations (Young et al., 2024). In Kelheim Germany a pilot operation of autonomous vehicles with no fixed route service is tested in public. The service is mixed with human drivers. Agent-based models calibrated on real demand data from the operation assessed several AV configurations and found that expanding the AV fleet can lead to higher efficiency in AV operations compared to conventional services (Schlenger et al., 2023). In summary, many DRT operations have been evaluated to better understand the factors contributing to their success and failure. These studies can inform future DRT operations and help with realistic expectations. Key insights reveal that effective service design tailored to community needs is essential while understanding user behavior—such as preferences for wait times over detour times—can enhance service satisfaction. Additionally, advanced planning and the potential integration of autonomous vehicles may significantly improve cost efficiency. Finally, the characteristics of the built environment are critical to DRT success, underscoring the need for thoughtful urban planning to facilitate effective DRT systems. This paper aims to add to these efforts through a comprehensive analysis of the system's performance and operational characteristics, along with a predictive model for the level of service based on demand, supply, and other factors. It also provides insights into customer behavior over time using innovative metrics from marketing literature, which can inform strategies to boost DRT usage and identify opportunities to enhance operational efficiency.

Background

The “GoSutton” DRT trial launched in May 2019 in the borough of Sutton, a suburb of the city of London, serving an area of 29 Km². The service, which was a partnership between Transport for London (TfL), ViaVan, and GoAhead, ran from 6:30 AM to 21:30 PM every day of the week. The trial was later ceased in March 2020, prior to its planned end date, due to the Covid 19 pandemic. Sutton is also well-served by local and intercity bus routes, as well as railways (9 local bus routes and 3 railway lines). The trial aimed at providing additional mobility options to the residents to move around the service area.

The trial offered curb-to-curb trips that were booked on-demand for

groups of up to 5 passengers per request. With no bookings in advance, passengers could only choose between a set of arrival time options for their trips.¹ The arrival time choices were calculated in real time, based on trip origin/destination and seat availability, and were communicated with passengers through the app. Depending on the demand and time of day 5–8 wheelchair accessible vans with 14-seat capacity served the requests. Phone bookings, accommodation of service animals, and free onboard Wi-Fi were also provided. Eligible users with disabilities could use a discounted pass for the trips (also known as freedom-pass owners). The trial had two phases, with the second phase starting ten weeks into the trial. The second phase was associated with an expansion of the area served (Fig. 1).

A change in the fare structure was also introduced in the second phase. Rides initially were charged a flat fee of £3.5. However, in the second phase, fares changed to better reflect the length and time of the trips, with longer trips and trips in the peak periods costing more. Table 1 shows the final fare structure.

General characteristics

Trip data were available for the duration of the service, amounting to almost 81,000 trips. The data contained information such as pick-up and drop-off times and locations of a served request, the time when a request was first made, and the fare that was charged.

Fig. 2 shows the evolution of daily ridership during the trial. The overall trend shows a gradual increase followed by a month of stabilized ridership, until it drops and later ceases due to Covid-19 concerns. The observed periodic pattern in the data follows a weekly surge with a lower number of trips on the weekends and higher numbers during the weekdays. The vertical red line marks the start of phase 2 of the trial, which coincides with an immediate jump in ridership. Expanding the service area and modifying the fare structure was clearly successful in attracting more riders. The analysis that follows focuses on the second phase only, as the majority of the trips occurred in that phase.

Fig. 3 depicts hourly demand levels as a function of the time of day, averaged over all days in the second phase for the corresponding hour. Fig. 3a shows the average number of trips at different times of the day for different days of the week. There is no distinct peak which indicates that trips are not following typical commute patterns, although trips are higher in the morning hours. Weekend trips are lower than weekdays. Fig. 3b shows the average number of simultaneous trips, in 15-minute intervals, served by the DRT service during the day, for different days of the week. Simultaneous trips are requests that are being served by any of the DRT vehicles at the same time interval. The number of requests that are on the move represents the number of simultaneous trips in those 15 minutes. Since the average number of simultaneous trips is smaller than the number of DRT vehicles, it is safe to assume that shared rides are not common. Early afternoon appears to be the period with the highest number of simultaneous trips although the number of requests in that period is lower than in the morning.

Further analysis of the travel patterns was conducted to better understand the absence of peaks in the hourly demand. The spatial and temporal gaps between consecutive trips of users during a day are plotted using the three-dimensional histogram in Fig. 4. A spatial (temporal) gap is the distance (time difference) between a rider's drop off and the pickup of the subsequent trip. The histogram illustrates that the majority of users spend 1–4 hours participating in activities at their destination which does not match the travel patterns of commuters and corroborates the findings in Fig. 3a.

¹ <https://moderngov.sutton.gov.uk/documents/s67709/Demand%20Responsive%20Bus%20-%20Go%20Sutton.Lite%20updated.pdf>

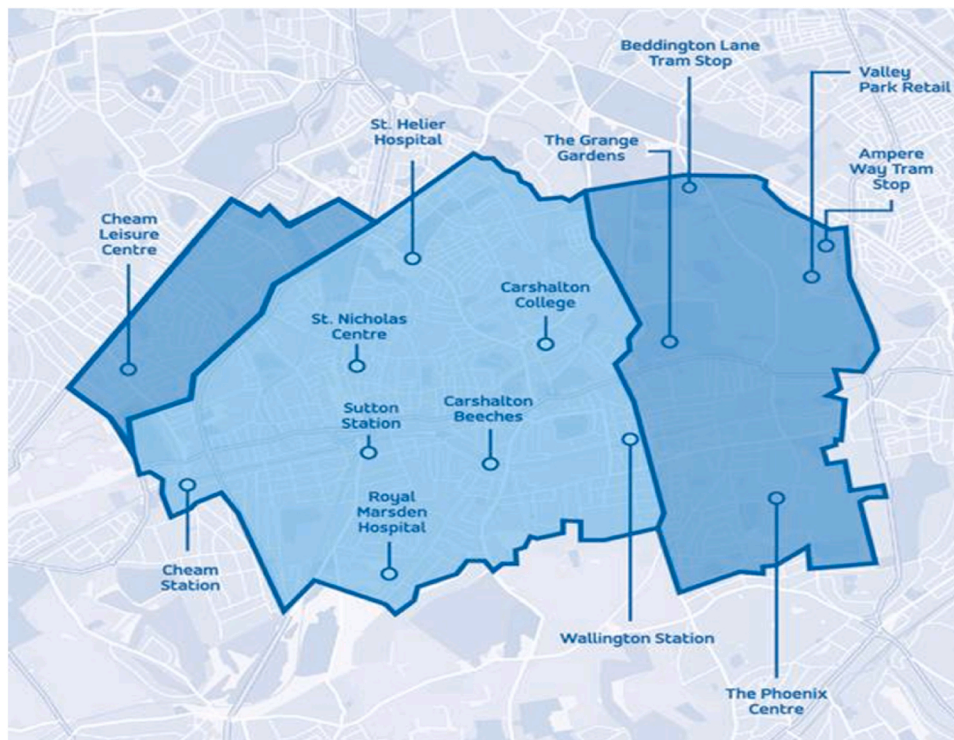


Fig. 1. GoSutton service area; before and after expansion. The expansion is marked by dark blue. (Source: TfL²¹).

Table 1
DRT service fare structure.

	Peak 6:30–9:30 am, 4–7 pm weekdays	Off-peak All other times including weekends and public holidays
Short rides ≤ 2 miles	£2.0	£2.0
Long rides > 2 miles	£3.5	£2.5

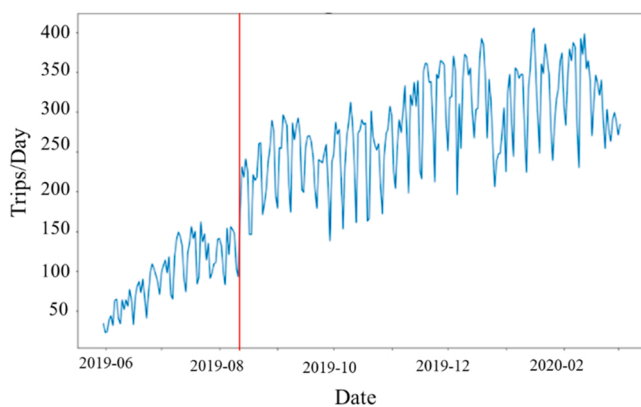


Fig. 2. DRT trial daily ridership.

Level of service analysis and models

In this section, we analyze the performance of the service to identify the factors that impact rider experience. Fig. 5 illustrates the LOS experienced by the riders in terms of the average ride times and wait times of trips as a function of time of day (in one-hour intervals). The wait times and ride times peak around 7:00 and the wait times peak once

more around 14:00.

To better understand the factors impacting the level of service a model is estimated to predict wait times and ride times using demand and supply variables and variables pertaining to the characteristics of the service and the region. The model can help planners understand what levels of service to expect in a DRT operation and how to improve it and is intended to help operators with medium-term decisions on fleet size and staffing. Generic DRT performance models exist in the literature. For example, Wilson and Hendrickson (1980) used simulation to develop empirical models that predict wait times and ride times using explanatory variables such as demand rate, number of vehicles, service area, and average vehicle speed. Daganzo et al. (Daganzo and Ouyang, 2019) developed an analytical model that uses similar variables.

The model is mainly focused on the characteristics of the trial under consideration to better understand what variables affect the performance of DRT services. Initial analysis of the data revealed a high (73 %) correlation between ride times and wait times. That is expected as high ride times may also result in high wait times and travel times. Therefore, a simultaneous regression model was used to predict them jointly. The three-stage least square method was used to estimate the coefficients of the simultaneous equations (Zellner and Theil, 1992). The unit of analysis is one hour of operation and the corresponding variables are average values for the corresponding hour. Tables 2 and 3 show the estimated model parameters.

Fig. 6 shows the scatter plots of the actual ride times and wait times with model predictions. The Root Mean Square Errors (RSME) for the ride time and wait time models 140 (s) and 180 (s) respectively, indicate good performance even at the disaggregate level of analysis (with the exception of a few uncharacteristically high ride and wait time values).

Table 2 summarizes the model specification and estimation results for the ride time ($R^2 = 0.49$). The coefficients are reasonable. Demand rates, number of DRT vehicles, day of week, and time of day, impact the average ride time experienced by riders. The negative sign of the coefficient for the number of DRT vehicles, suggests that running an adequate number of vehicles is important as it can decrease the ride times and increase LOS. Rides take more time on weekdays compared to

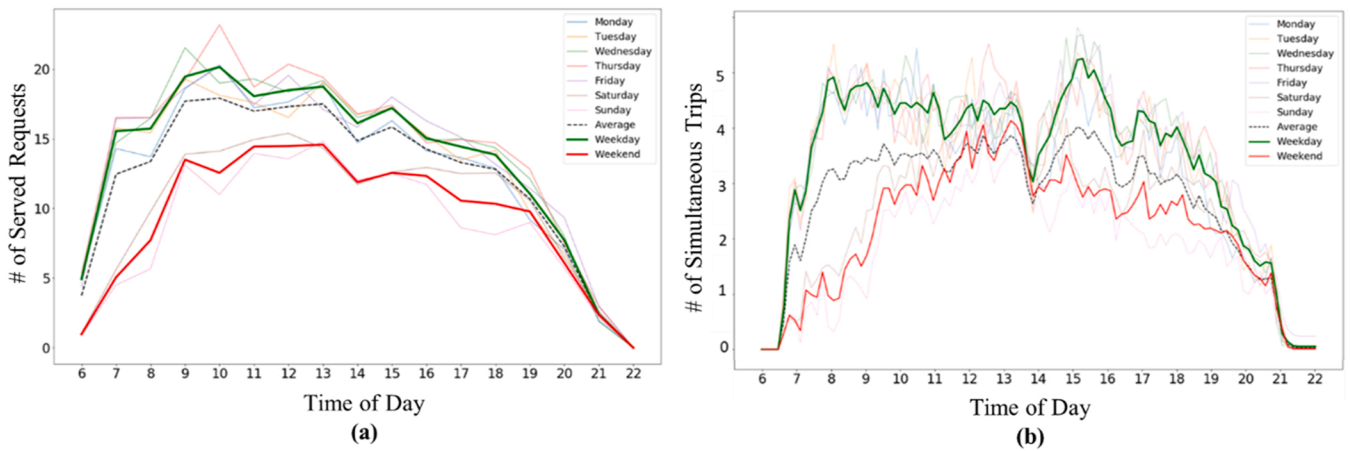


Fig. 3. Demand levels for different days of the week and times of the day. (a) Average hourly demand. (b) Average number of simultaneous trips in 15-minute intervals.

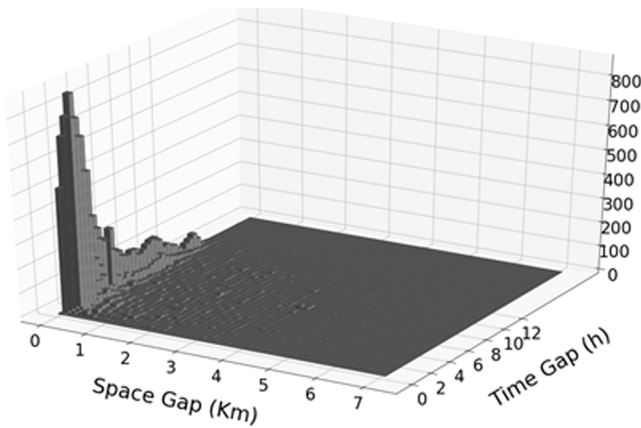


Fig. 4. The spatiotemporal gap between consecutive trips of users.

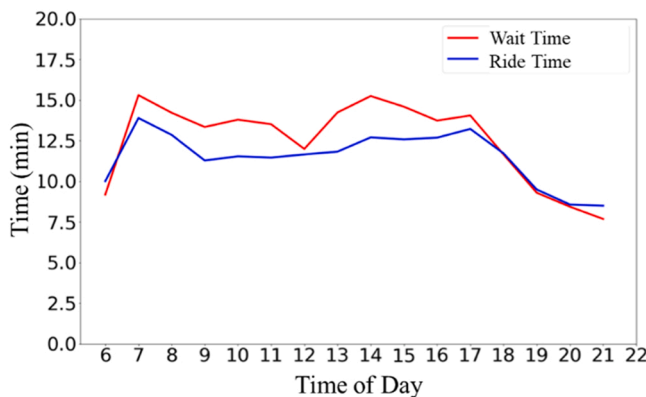


Fig. 5. Level of service by time of day.

weekends. The time-of-day variables capture the traffic conditions in the area by having higher values for the morning and evening peak periods. The impact of time of day on the LOS offered by DRT services indicates that the number of required vehicles during these periods should be properly determined to compensate for the impact of time of day to meet

Table 2

Model specifications for predicting hourly ride time 9 The number in parentheses are t-statistics).

Variable	Coefficient (t-statistics)
Intercept	530 (20.8)
# Of DRT vehicles, N_{DRT}	-40 (-7.0)
Demand rate, D (trips/hr)	9 (10.0)
Dummy for the weekday, W	113 (11.9)
Dummy for time of the day, X_{6-7}	48 (2.0)
Dummy for time of the day, X_{7-8}	294 (10.3)
Dummy for time of the day, X_{8-9}	285 (10.1)
Dummy for time of the day, X_{9-10}	215 (7.4)
Dummy for time of the day, X_{10-11}	198 (6.8)
Dummy for time of the day, X_{11-12}	217 (7.6)
Dummy for time of the day, X_{12-13}	265 (9.0)
Dummy for time of the day, X_{13-14}	237 (8.3)
Dummy for time of the day, X_{14-15}	310 (11.6)
Dummy for time of the day, X_{16-17}	370 (13.0)
Dummy for time of the day, X_{17-18}	371 (13.9)
Dummy for time of the day, X_{18-19}	342 (13.0)
Dummy for time of the day, X_{19-20}	121 (4.3)
Dummy for time of the day, X_{20-21}	107 (3.9)

Note that the time of day variables X_{i-j} represents the time between hour i and j in a day starting from midnight. The base variable is the time between hours 21 and 22.

Table 3

Model specification for predicting average wait time (The number in parentheses are t-statistics).

Variable	Coefficient (t-statistics)
Intercept	-432 (-13.7)
Average wait time, Y_{Wait}	1.46 (34.7)
Ratio of discounted pass owners, (RDP)	340 (7.1)

a target LOS objective.

Table 3 summarizes the model specification and estimation results for average wait times ($R^2 = 0.60$). The model uses the average ride time as an explanatory variable. As expected, larger values of average ride times contribute to larger values of average wait times. The ratio of passengers who booked a trip with a discounted pass over all passengers in the one hour of operation is an important variable (ranging from 0 to 1). The positive sign for the coefficient corroborates the fact that riders using discounted passes, typically associated with passengers with disabilities, require larger dwell times for pick up and drop off which can increase the wait time for other passengers. The model suggests that

² <https://moderngov.sutton.gov.uk/documents/s67709/Demand%20Responsive%20Bus%20-%20Go%20Sutton.Lite%20updated.pdf>

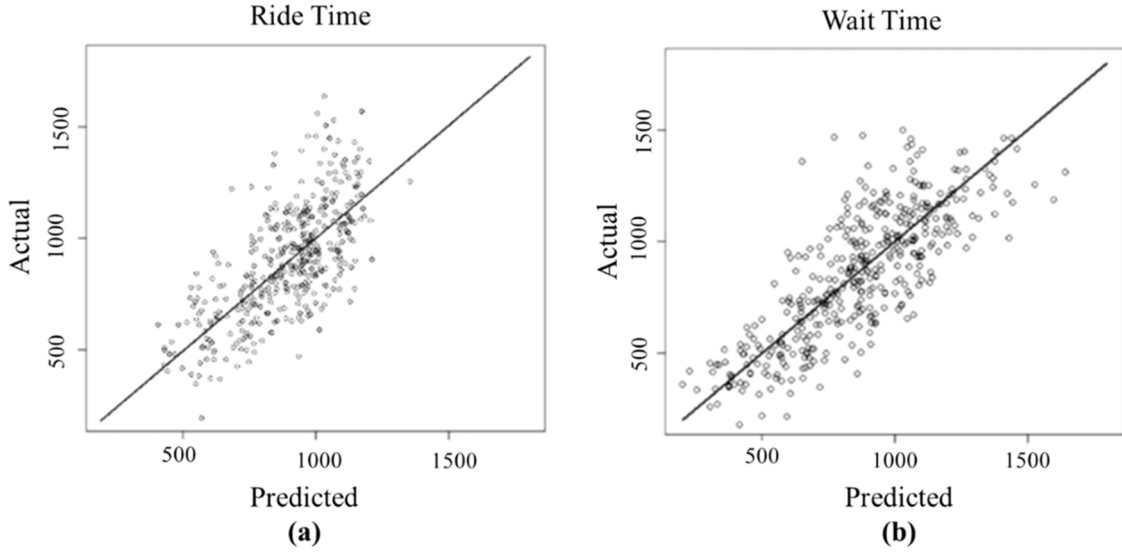


Fig. 6. Predicted vs actual values (s). (a) ride time, (b) wait time.

countermeasures should be employed to avoid LOS deterioration during the times when discounted pass users constitute a large share of booked rides. Examples of such measures include the use of a larger number of vehicles, during periods with high volumes of discounted pass users, better training of operators, etc.

In addition to the insights discussed above, the LOS model can help operators adjust their supply with demand and provide more realistic information to users with respect to their expected pick-up time and arrival time at their destination. Note that the explanatory variables are selected to help operators with medium-term decisions on fleet size and staffing, relying on information they have readily available from bookings. However, other variables and model structures with higher predictive power can be considered to improve prediction accuracy. For example, trip distance has a relationship with ride time. Traffic and weather conditions and speed data can be used instead of time-of-day variables. Also, more advanced Machine Learning models can be used to better predict LOS variables on a more granular level. For example, the average ride time can be broken down into its major components such as the direct distance, detour, and delays. Finally, the competition between alternative modes can also be considered.

Passenger loyalty analysis

Typical studies of passenger behavior focus on analyzing travel patterns in terms of time, location, and trip purpose. In this study, we explore how customers make purchase decisions with regard to the DRT service to understand how techniques used by marketing campaigns in the general consumer space can be used to promote the service more effectively. A longitudinal analysis of customer interactions with the DRT service, as well as recommendations on how to promote their usage is presented in this section.

The behavior of customers is analyzed through measures of customer loyalty. From a marketing point of view, loyalty is the user's desire to continue their relationship with a particular business for a long time. There are various Key Performance Indicators (KPIs) that measure customer loyalty (ISORAITÉ., 2016). Chances of return, cross-sell ratio, and re-purchase ratio are used for this analysis. A definition of each KPI is provided below:

Chances of return: This KPI measures the probability of the return of a customer who already bought a certain number of products for more purchases. In the context of this paper, this is the probability that a user will use the service more, given that they have already used it n times.

$$CR_x = \frac{N_{x+1}}{N_x} \quad (3)$$

where CR_n is the chance that a customer who already made x purchases will make an additional purchase later on. N_x is the number of customers who made x purchases.

Cross-sell ratio: This KPI assesses the attractiveness of a service to its customers by measuring the ratio of customers who purchased new and different products (Murray, 2019). Cross-sell ratio is defined as the percentage of a cohort who purchased another product. A cohort is a group of customers who made a purchase during a given week (or other time intervals of interest). In the context of this paper, a different product may represent a trip during the weekend, or in a different time period. The cross-sell ratio for a cohort of customers can be calculated using Eq. 4. A cohort is a group of customers that made a purchase during a certain time period.

$$CSR_i = \frac{N'_i}{N_i} \quad (4)$$

where CSR_i is the cross-sell ratio of the cohort i , N_i is the number of customers in a cohort i , and N'_i is the number of customers in a cohort i who also purchased a different product later on.

Re-purchase ratio: This KPI addresses the attractiveness of a service to its customers by measuring the ratio of customers who purchase a product again. The re-purchase ratio is defined as the percentage of a cohort having placed another order within a certain period of time (e.g., a week, a month, etc.) (Sutto, 2016). The re-purchase ratio can be calculated using Eq. 5.

$$RPR_i^T = \frac{N_i^T}{N_i} \quad (5)$$

where RPR_i^T is the re-purchase ratio of the cohort i measured for time period T , N_i is the number of customers in a cohort i , and N_i^T is the number of customers in a cohort i who made another purchase (trip) within time period T measured after their first purchase (trip).

Fig. 7 plots the breakdown of new and old customers and the aforementioned KPIs over time on a weekly basis. Fig. 7a focuses on a cohort of riders who purchased a trip in a given week and plots the number of new riders in that week as well as the number of riders that used the service in previous weeks. The drop around the 24th week is due to the Christmas and New Year holidays. The figure does not show

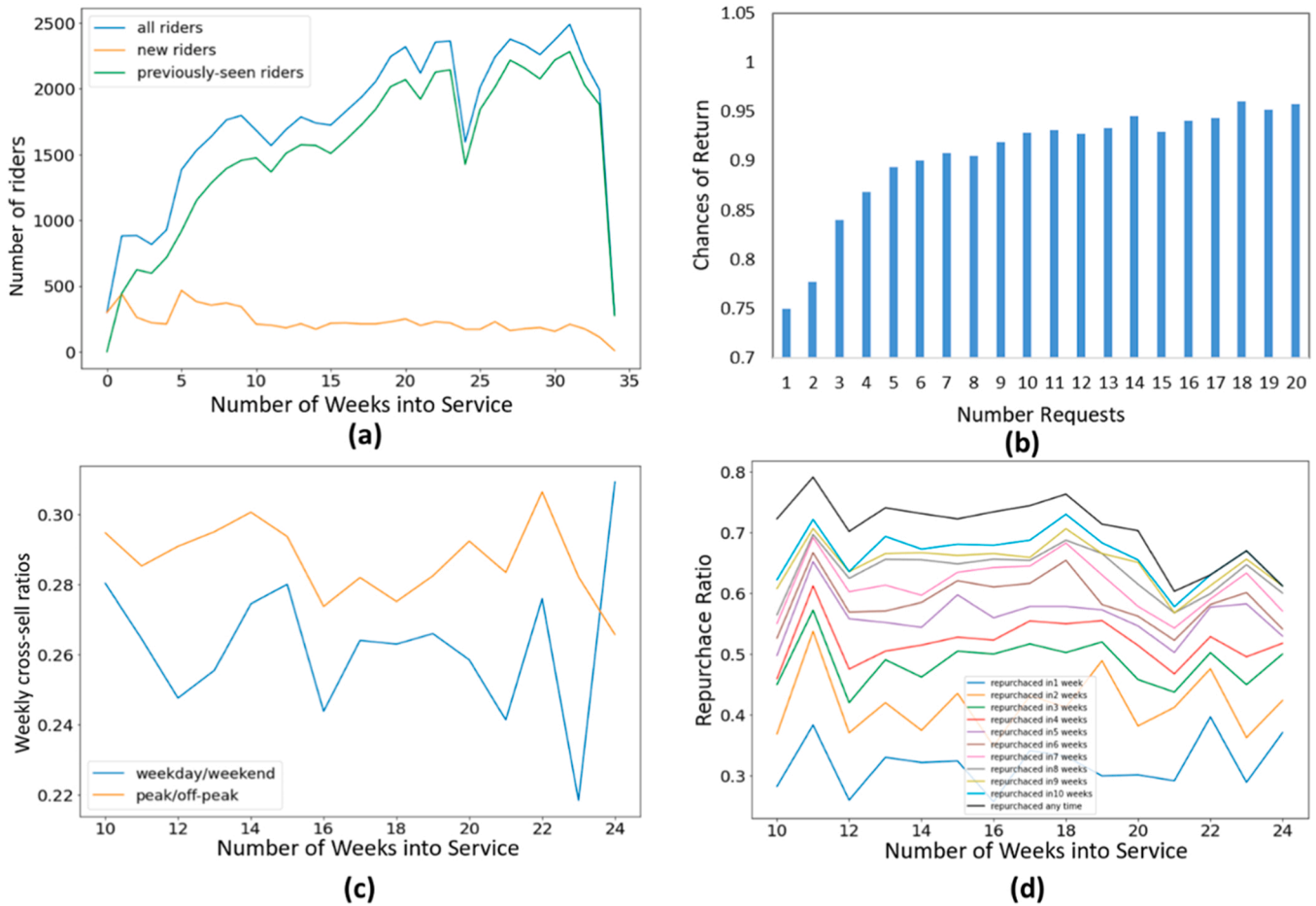


Fig. 7. Passenger loyalty KPIs. (a) Weekly new and old riders. (b) Chances of return for a customer who purchased a certain number of trips. (c) Weekly cross-sell ratios. Peak/off-peak and weekday/weekend trips are considered as cross-purchases. (d) Re-purchase ratios for various time periods.

the typical curve of adoption of a new product as the number of new users does not increase over time. A typical adoption curve is S-shaped. It is expected that as more customers use the service, the word of mouth about the service would result in attracting more and more customers until demand hits a saturation point. However, in this case, the downward slope of the line representing new customers indicates that the service could benefit from better promotions and marketing strategies. Referral promotion is an example of such strategies that have proven effective in various contexts. Many app-based services, for example, from e-commerce to e-hailing, provide discounts to users who refer the service to a new user through the app. The discount can be applied when the new user either creates an account or makes the first purchase. DRT services can deploy similar strategies, especially at the early stages of a field test to attract new riders at an increasing rate.

The figure also demonstrates that the majority of trips are made by riders who have used the system in the past. This highlights the importance of analyzing the behavior of “loyal passengers”. Fig. 7b shows the probability of purchasing more trips for customers who have made at least a certain number of trips in the past. The figure indicates that the more customers purchase trips, the more likely they are to purchase again in the future. For example, 75 % of customers who made one purchase ended up making at least one more purchase, whereas almost 90 % of riders who made at least 5 purchases, also used the service later on. The graph is suggesting that, in addition to the referral promotions, the DRT service could also benefit from promotions that provide discounts on the first few rides of a new rider. These promotions encourage the users to both try and familiarize themselves with the service. Such strategies are commonly used by e-hailing companies such

as Uber and Lyft.

Fig. 7c shows the cross-sell ratio KPI for different weeks of the service (the first/last 10 weeks are omitted to avoid boundary effects). In this analysis, trips in the peak and off-peak periods, as well as trips in the weekends and weekdays are viewed as different products. The cross-sell ratios show that 28–30 percent of riders who purchased a trip during the peak period also purchased at least one during the off-peak. The same is true about weekdays and weekends. 22–30 % of passengers who purchased a trip on a weekday purchased at least one during the weekend. The cross-sell ratios suggest that the DRT service could benefit from offering bundles and passes that provide a collage of trips with different characteristics. The bundles can be designed to provide tailored prices aiming at increasing ridership.

Fig. 7d shows the re-purchase ratio KPI for cohorts of users having their first trips in the same week. The horizontal axis shows the number of weeks into the service and represents the cohorts who made their first trips in those weeks. The vertical axis is the re-purchase ratio and each trend line is a time period for which the repurchase ratio is calculated. As an example, a passenger may not make a second purchase during the first two weeks after its first purchase but end up making the second purchase in the third week. The graph shows a discernable difference in the re-purchase ratio of cohorts having their first trips in some weeks as opposed to other weeks. For example, passengers who used the service for the first time in weeks 12 and 21 have lower re-purchase ratios for any time period compared to their adjacent weeks (i.e. they are less likely to use the service again). On the other hand, week 11 has higher re-purchase ratios than any other week. The average wait times and ride times do not change much from week to week and therefore, LOS may

not explain the observed difference. In any case, the KPI is useful for monitoring conditions. It can alert the service operators to have a closer look at those weeks, and review customer feedback, ratings, and level of service to better understand the reasons behind this difference. Operators may also contact the users of the week of concern, offering explanations for the lower LOS (if they were performance issues during the corresponding week).

Complementarity with paratransit

Many DRT trials suffer from the presence of other, similar modes of travel and parallel services that run in conjunction with the trial. These parallel services cause inefficiencies and limit the benefits that come from economies of scale. For example, Centennial Innovation Team suggests that the operation of “call-n-ride” and “Access-a-ride” in the same area may have contributed to the low ridership of the “Go Centennial” DRT program in Colorado (Team et al., 2017). In Sutton, a paratransit service also operated in the area providing door-to-door trips to eligible users. The paratransit service required in-advance bookings (one day prior) and was only available for non-commute trips. The service is available to riders who have proven disabilities. Eligible paratransit users were also eligible for the discounted pass offered by the DRT service. Given that more than 40 % of DRT trips were made by riders with discounted passes, a large overlap may exist between the users of the two services. In this section, we evaluate a system where the trips of both services are pooled and served with one unified DRT service. The consolidation can improve efficiency, increase shared rides, and reduce required fleet size while maintaining a target level of service. While in actual applications it may not be practical to combine the paratransit and DRT services for all trips, the analysis can be used to assess the overall potential of such consolidation.

The potential of a consolidated DRT service as opposed to the independently operated DRT and paratransit services, is evaluated assuming all requests can be pooled together and served by one service. Paratransit data for three consecutive days during the DRT trial period that took place in the morning peak (7–8:00), evening peak (16–17:00), and midday off-peak (10–11:00 AM) were used. On average, there were 13 paratransit trips/hr. The paratransit requests were pooled together with the DRT requests for the corresponding hour. Subsequently, all requests were scheduled using a Dial-a-Ride (DaR) formulation of the problem. The DaR problem has been extensively studied in the literature and deals with the routing and scheduling of a set of vehicles to serve a set of pickup and delivery requests between origins and destinations (Cordeau and Laporte, 2007). The aim is to plan a set of vehicle routes, under LOS constraints, to serve the requests with minimum cost.

The DaR problem was solved three times for each hour: 1. DRT trips only; 2. paratransit trips only; and 3. combined DRT and paratransit trips. The minimum required fleet size and Vehicle Kilometers Traveled (VKT) for each case are measured and compared. The following assumptions were made:

1. All requests are known in advance
2. Vehicles have a capacity of 10 seats. They start and end their runs at a depot
3. The maximum allowed wait time of a rider (the difference between actual pickup time and desired pickup time) is 10 minutes
4. The maximum allowed in-vehicle ride time of a rider is 1.25 times the direct travel time of the same trip
5. The DaR aims at minimizing the VKT, i.e. the objective function is the sum of all kilometers traveled by the vehicles

Table 4 summarizes the solution of the DaR problem for each scenario for different days of the week and different time periods. The formulation proposed in (Cordeau and Laporte, 2007) was used to obtain the corresponding solutions. The results show an overall decrease in the total VKT when paratransit and DRT trips are operated jointly. Except for one case, all other cases have lower VKTs in the consolidated operations. An average of a 13 % decrease in VKTs was observed across all cases, and in the one case of increase in VKT, the fleet size is reduced, and therefore, there are overall savings. In 5 out of 9 test cases, the joint operation of the two services reduced fleet size, i.e., it is possible to serve the trips with a smaller number of vehicles if the trips of both services are pooled into one service. On average, the required fleet had 0.78 fewer over the 9 test cases.

The results are promising and indicate that service operators should seek opportunities for economies of scale through co-operation with other services. Although, in practice the integration of general DRT and paratransit services is more complicated, there is potential for benefits that should at least be explored.

Conclusion and discussion

This paper presented a comprehensive analysis of the GoSutton demand-responsive transportation (DRT) trial conducted in the borough of Sutton, London. The trial’s evaluation provides insights into the performance of DRT services, revealing critical factors that influence service levels, efficiency, and ridership.

Using granular trip data, the paper discusses the general characteristics of the operation and its users and depicts spatio-temporal features of the trips. Results show an increase in the service area coupled with a decrease in fare prices boosted the ridership almost immediately. Also, the timing of trips and the time spent at destinations show that most trips were taken for non-work purposes.

By developing a performance model, this study elucidates the interplay between demand, supply, operations characteristics, and LOS, offering a framework that can inform future DRT deployments. While other research generally focused on built environment and the impacts of having passengers on board on LOS, the model developed in this study underscores the significance of time-of-day variations and customer composition on DRT performance. The model reveals the need for planning extra capacity and possible deterioration of wait times if a

Table 4
Comparison of the consolidated and independent paratransit and DRT operations.

		Independent Operations				Consolidated DRT & Paratransit		Savings	
		DRT		Paratransit		VKT (km)	Fleet Size (# of veh)	VKT (km)	Fleet Size (# of veh)
		VKT (km)	Fleet Size (# of veh)	VKT (km)	Fleet Size (# of veh)				
Tuesday	Morning	63.76	4	71.34	6	127.10	9	6 %	10 %
	Midday	100.69	6	95.75	4	164.92	10	16 %	0 %
	Evening	88.15	6	94.57	4	120.60	8	34 %	20 %
Wednesday	Morning	87.84	6	19.25	4	113.08	9	−6 %	10 %
	Midday	72.66	6	68.72	4	90.36	8	36 %	20 %
	Evening	97.90	6	5.93	2	99.76	8	4 %	0 %
Thursday	Morning	114.18	8	33.39	2	121.95	10	17 %	0 %
	Midday	105.59	6	85.50	4	187.30	10	2 %	0 %
	Evening	56.85	4	75.36	4	121.24	7	8 %	13 %

large ratio of customers needs additional dwell times for boarding and alighting.

A longitudinal analysis of customer behavior from a marketing perspective suggests that GoSutton could enhance its ridership through targeted promotional strategies. The results showed that most users that made one trip continued using the service, therefore strategies such as referral promotions that incentivize existing users to introduce new ones could have a great impact. Other recommended strategies include introductory discounts for new users, and bundling passes that offer trips at various times and for different day types. This type of analysis and recommendations are novel for DRT services to the best of our knowledge but resonate with broader trends observed in ride-hailing apps worldwide.

Additionally, the study explored the potential for operational efficiency improvements by integrating GoSutton with existing paratransit services providing additional insight on how to better optimize DRT services. This approach suggests that consolidating services can lead to significant benefits, including fleet size reductions and improved vehicle kilometers traveled (VKT). The paper uses trip data from both operations and demonstrates that combining the operations can on average decrease VMT and fleet size fleet size by 13 % and 8 percent respectively.

Overall, the GoSutton trial provides a valuable case study for understanding the dynamics of DRT operations and offers actionable insights for enhancing service delivery. By focusing on demand patterns, customer engagement strategies, and service integration, the findings offer a pathway for improving DRT services' effectiveness and sustainability. Future research and practice should build upon these insights, exploring further integration opportunities and innovative strategies to advance the field of demand-responsive transportation.

The ideas presented in this paper are generalizable for future deployments of DRT services, however, the details of each example may only apply to GoSutton trial. For example, the importance of time of day in LOS model may decrease or increase as traffic conditions and vehicle speeds vary across areas. Also, not all DRT services provide discount rides for passengers with special needs. Additionally, the successful integration of DRT services with paratransit ones depends on the spatial distribution of the two operations. In the case of GoSutton paratransit trips overlapped significantly with the service area making it a great candidate for service integration.

CRedit authorship contribution statement

Sayedmostafa Zahedi: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Zhenliang Ma:** Writing – review & editing, Supervision, Investigation, Formal analysis. **Haris N. Koutsopoulos:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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